Generating ideas

Name: Unit:

1 I write down one idea that works.
Point to the idea and to what makes it work.

2 I write down several ideas that work in different ways, not small changes of the same idea.
Point to each idea and to what makes it different from the others.

3 I sort my ideas into kinds and add one from a kind I had missed.
Point to your groups, their names and the idea you added to fill the gap.

4 I pick the idea that fits best when one condition changes, explain why, and say how sure I am.
Point to the changed condition, the idea you picked, your reason and the words that say how sure you are.

The ladder is for students' own self-check. It is never graded. Circle the step your answer reached.

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Which step did my answer reach?	What in my answer shows it?	My next step
1 · 2 · 3 · 4		

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Evaluating and choosing

Name: Unit:

1 I make a choice or a judgement and give one reason for it.
Point to your choice and the one reason.

2 I give several reasons, each backed by a fact or a rule I can check.
Point to each reason and the fact or rule beside it.

3 I weigh my reasons against each other and against the other side, and show which one decides it.
Point to where you answer the other side and the sentence that says which reason weighs most.

4 I test my judgement on a new case or a new piece of evidence, and say how sure I am.
Point to the new case or evidence, whether your judgement held, and the words that say how sure you are.

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Comparing and transferring

Name: Unit:

1 I name one thing the two cases share.
Point to the shared feature in both cases.

2 I name several things that match and at least one place where the cases differ.
Point to each match and to the difference.

3 I explain the rule that makes the cases match, and where the comparison breaks.
Point to the sentence that states the rule and to the place where the comparison breaks.

4 I use the rule on a new case, and say how sure I am that it fits.
Point to the new case, your rule used on it, and the words that say how sure you are.

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Explaining and reasoning from basics

Name: Unit:

1. I give one explanation or one fact I know for certain.
   Point to the explanation or the fact.
2. I give several explanations or steps, and say what each one rests on.
   Point to each explanation or step and what it rests on: evidence, a known fact or an assumption.
3. I link my steps into one line of reasoning, or show which explanation fits the evidence best.
   Point to the linking words (because, so, which means) or to the evidence that rules out the other explanations.
4. I use my reasoning on a new case or a changed number, say where it stops working, and say how sure I am.
   Point to the new case, the place where your reasoning stops working, and the words that say how sure you are.

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Causes and systems

Name: Unit:

1. I name one cause and one effect.
   Point to the cause, the effect and the word that links them.
2. I build a chain: A leads to B leads to C.
   Point to each link in the chain: at least three parts in a row.
3. I find a loop: C affects A again, or a delay or a side effect.
   Point to the arrow or sentence that leads back to the start, or that names the delay or side effect.
4. I use the loop to predict what happens if one part changes, and say how sure I am.
   Point to the part you changed, your prediction and the words that say how sure you are.

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Regulating your own thinking

Name: _____ Unit: _____

1. I name one thing I did while I worked.
   Point to the strategy you name, such as "I drew it". "I thought harder" does not count.
2. I describe several steps in how I worked, and one moment that shaped how I worked, such as changing my approach or getting stuck.
   Point to each step and to the moment that shaped how you worked.
3. I explain what made me change my approach or moved me on, and why it helped.
   Point to the sentence that says what made you change your approach or moved you on, and why it helped.
4. I say what I would change next time and why, and how sure I am that it would help.
   Point to what you would change, your reason and the words that say how sure you are.

Students rate the ladder on the Think back part at the end of each task, not on the task answer.

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